Evaluation 2: Duolingo XR — Immersive Chinese Learning

YINING ZHU
49053881

Objective and Validation Metrics

Interactive Prototype 2a extends the previous Duolingo XR prototype from simple object-based vocabulary learning into a more complete VR learning experience. In the previous prototype, users mainly explored a virtual Study Room, clicked objects, viewed vocabulary cards, and listened to Chinese pronunciation. Evaluation 1 showed that connecting vocabulary with 3D objects was useful, but users needed clearer onboarding, more physical interaction, and stronger feedback.

At the beginning of the experience, participants learned five target words: 书 (shū, book), 桌子 (zhuōzi, desk), 电脑 (diànnǎo, computer), 找到 (zhǎodào, find), and 拿起 (náqǐ, pick up)**.** Each learning card presented the Chinese characters, Pinyin, English meaning and pronunciation audio. After completing this learning stage, participants entered the virtual room, where they could explore and interact with objects before beginning two Chinese-only tasks: 找到电脑 and 拿起书**.** This structure was designed to move from supported vocabulary learning to less-supported application in the VR environment.

The purpose of Evaluation 2 was to investigate whether beginner learners could move from supported vocabulary learning to understanding and performing simple Chinese instructions in VR. The evaluation focused on six questions:

- Can users remember the five target Chinese words after the learning stage?
- Is the Explore Room stage useful as a vocabulary review activity?
- Can users understand simple Chinese-only task instructions without English translation?
- Can users distinguish between selecting an object and physically grabbing it?
- Does the system provide enough feedback for users to understand their progress?
- Does physical interaction make the Chinese-learning experience more engaging?

Five participants took part in the evaluation. Three had no previous Chinese-learning experience and two had a small amount of Chinese experience. Participants also had different levels of previous VR experience. This helped identify whether difficulties came from Chinese understanding, VR interaction, or the prototype itself.

Success was defined as most participants completing the learning stage independently, at least four out of five participants completing the main tasks, at least four participants recalling three or more target words, and average ease and enjoyment ratings of at least 3/5.

Results

Overall, participants were able to use the learning stage successfully and showed some immediate vocabulary recall. The average post-test vocabulary recall was **3.6 out of 5 words**. Participants with some previous Chinese experience generally remembered more vocabulary, while complete beginners showed more difficulty with action words.

Participant	Chinese exp.	VR confidence	Learning time	Task 1 first try	Words recalled
P1	None	2/5	2:21	No	3/5
P2	None	4/5	1:48	Yes	4/5
P3	None	3/5	1:35	No	2/5
P4	Basic	4/5	1:39	Yes	4/5
P5	Little	3/5	1:02	Yes	5/5

On average, participants remembered 3.6 out of 5 target words. The average enjoyment rating was 4 out of 5, while the average ease rating was 3.4 out of 5. Participants with some previous Chinese experience generally completed the tasks more confidently and remembered more vocabulary. Concrete nouns such as 书 (book) and 电脑 (computer) were easier to remember than action words such as 找到 (find) and 拿起 (pick up). Three participants correctly completed Task 1 on their first attempt. However, a significant prototype issue arose during testing: after participants selected the computer, they received no feedback indicating whether their choice was correct or incorrect, and Task 2 did not appear automatically. As a result, I had to manually explain the instructions for Task 2 to all participants.

The Explore Room was helpful for reviewing vocabulary, but only three items were accompanied by vocabulary cards, and these cards did not include audio pronunciations. Some participants noticed this and expected all learning items to function in the same way.

Analysis

Insight 1 — Missing feedback affected both gameplay and the quality of the evaluation

The largest usability problem was not the Chinese instruction itself but the lack of system response. After the computer was selected, the prototype did not show “Great job” / “Try again” and did not advance to “拿起书”. This made correct actions look identical to incorrect or unregistered actions. As a result, Task 2 required facilitator intervention and the intended autonomous mission flow was broken. This also weakens the validity of the language evaluation: when the system does not confirm a selection, it is difficult to distinguish whether hesitation comes from Chinese comprehension, controller uncertainty, or a technical state problem. The next iteration must therefore prioritise reliable task-state logic before adding more learning content.

Insight 2 — VR was present, but not yet central to learning

Although Prototype 2a introduced a VR headset, controllers and physical interaction, much of the learning experience still followed the same pattern as the previous prototype: learners read vocabulary information and selected an object. This suggests that the concept had moved into VR, but VR was not yet fully used as the learning mechanism. The most engaging moments occurred when participants physically interacted with objects rather than only reading cards. Therefore, the next iteration should use actions such as Grab, Move and Place as part of the language-learning activity itself.

Insight 3 — Physical actions have strong potential for teaching action vocabulary

The “拿起书” task moves the concept closer to the original XR goal because learners must connect a Chinese action word with a physical Grab interaction. The current testing record also shows Select-versus-Grab hesitation among lower-confidence VR users. Therefore, failure on this task can reflect controller knowledge rather than Chinese knowledge. A short controller tutorial should happen before the language assessment. The learning card for action words should also use a simple animation or gesture (for example, a hand lifting a book) so that the learner sees, hears and performs the same action. This would make the embodied-learning rationale more explicit.

Insight 4 — Learning support should be reduced gradually

The testing also showed that removing too much support immediately after the vocabulary-learning stage was difficult for some beginners. Participants with no Chinese experience often depended on English, while participants with some Chinese knowledge were more comfortable using Pinyin. Action words were particularly difficult when only Chinese characters were shown. The next iteration should therefore use progressive support rather than moving directly from full translation to Chinese-only instructions. During the initial learning stage, cards can show Chinese, Pinyin, English and audio. During free exploration, the card can show Chinese, Pinyin and audio but remove English. During the task stage, Chinese can be shown first, with Pinyin or a visual hint only if the learner is having difficulty. This would create a smoother transition from supported learning to independent use.

Evaluation of Aims

The testing aims were partly validated. Participants could learn and remember some of the target vocabulary, especially concrete nouns such as 书 and 电脑, while action words such as 找到 and 拿起 were more difficult. The learning cards and pronunciation audio were generally effective, but support was inconsistent during room exploration because only some objects had review cards and no pronunciation replay was available. Chinese-only task instructions were also only partly validated. Some participants understood the instructions, while others relied on object recognition, guessing or trial-and-error. In

addition, the broken Task 1 → Task 2 transition made it difficult to judge language understanding accurately.

Overall, the evaluation supports continuing with VR physical interaction, but clearer feedback, more consistent learning support and more meaningful action-based tasks are needed in the next iteration.

Concept Iteration

Based on the evaluation, the next iteration will develop the prototype into a Chinese Room Tidying Challenge. The current prototype still relies heavily on vocabulary cards and simple object selection, while the testing showed that physical interaction has more potential to make the learning experience engaging. Therefore, the next version will connect Chinese vocabulary more directly with actions that are natural in VR, rather than simply adding more words or cards.

During free exploration, more everyday objects in the room will become optional learning objects, such as books, pens, chairs, lamps and drawers. Selecting an object will show its Chinese word, Pinyin and pronunciation audio, with English available as an optional hint. This allows learners to explore and learn more vocabulary at their own pace without requiring them to memorise every word. The main tasks will then focus on a smaller set of target vocabulary so that the learning difficulty remains manageable for beginners.

The main tasks will focus on a smaller set of target words and use Grab, Move and Place interactions. For example, learners may complete tasks such as 拿起书 (Pick up the book), 把书放在桌子上 (Put the book on the desk), and 把笔放进抽屉 (Put the pen into the drawer). This allows action words to be connected directly with physical movement. The next version will therefore shift from simply learning vocabulary in VR to using Chinese to complete physical tasks in VR.

Reflection and Next Steps

This evaluation was useful, but the testing process was less consistent than intended because the full task flow was not functioning reliably. The intended task flow was not fully working during testing. In particular, Task 1 did not provide clear feedback and Task 2 did not appear automatically, so I sometimes had to intervene and give participants the next instruction manually. This made the testing less consistent and also made it harder to distinguish whether a problem came from Chinese understanding, VR interaction, or the prototype itself.

I also realised that although the prototype had moved into VR, the learning experience still relied too much on vocabulary cards and simple object selection. This made the experience feel more like a vocabulary exercise inside a VR room rather than a fully interactive VR learning game. In the next iteration, I want to make the learning experience less repetitive and more engaging by using more natural VR interactions such as Grab, Move and Place. These actions can be connected directly with Chinese verbs and instructions, so the interaction itself becomes part of the learning.

For the next prototype, I will first improve the task flow and feedback so that the full experience can be tested without facilitator intervention. I will then develop the Chinese Room Tidying Challenge, where learners use Chinese instructions to pick up, move and place objects. In the next round of testing, I hope to run a more complete and consistent evaluation and better understand whether these physical interactions make the learning experience clearer, more enjoyable and less repetitive.

GitHub Link

https://github.com/yenyong1211/YININGZHU_DECO7230_2026/tree/main/Evaluation_2

https://github.com/yenyong1211/YININGZHU_DECO7230_2026/tree/main/Prototype2a

Appendix

Appendix A — Participant 1 Raw Data

Item	Result / Response
Chinese experience	None
VR experience	None
VR confidence	2/5
Vocabulary Learning	2 min 21 sec
Explore Room	1 min 5 sec
Task 1	37 sec
Task 2	29 sec
Core task total	**4 min 32 sec**
Difficult words	找到, 拿起
Words remembered	书, 电脑, 桌子 — **3/5**
Task 1 first attempt	No — selected Desk first
Meaning of 找到电脑	“Something about the computer, maybe find/select it.”
Meaning of 拿起书	“Pick up the book.” — correct after hesitation
Task 2 interaction	Clicked the book before grabbing; successful on second attempt
Most confusing part	Knowing when to click and when to grab; no feedback after Task 1
Most enjoyable part	Interacting with objects instead of only using flashcards
Suggested improvement	Show controller controls; clear task feedback and transitions; add audio to room cards
Ease	3/5
Enjoyment	4/5
Key observation	P1 used audio frequently and found nouns easier than action words. They needed help finding START TASK. After selecting the correct computer, no system response appeared, so they were unsure whether Task 1 had finished.

Appendix B — Participant 2 Raw Data

Item	Result / Response
Chinese experience	None
VR experience	Some
VR confidence	4/5
Vocabulary Learning	1 min 48 sec
Explore Room	47 sec
Task 1	12 sec
Task 2	19 sec
Core task total	**3 min 6 sec**
Difficult words	桌子, 找到
Words remembered	书, 电脑, 找到, 拿起 — **4/5**
Task 1 first attempt	Yes
Meaning of 找到电脑	“Find the computer.”
Meaning of 拿起书	“Pick up the book.”
Task 2 interaction	Grabbed successfully without first clicking
Most confusing part	Learning cards felt separate from the game; Task 1 had no completion feedback
Most enjoyable part	Using Chinese words to perform actions in VR
Suggested improvement	Fix task transition; add consistent cards and audio in the room
Ease	4/5
Enjoyment	4/5

Key observation	P2 understood the UI quickly and used Explore as a vocabulary review stage. They selected the computer correctly immediately, but waited because the prototype gave no response.

Appendix C — Participant 3 Raw Data

Item	Result / Response
Chinese experience	None
VR experience	Little
VR confidence	3/5
Vocabulary Learning	1 min 35 sec
Explore Room	1 min 9 sec
Task 1	36 sec
Task 2	34 sec
Core task total	**3 min 54 sec**
Difficult words	找到, 拿起, 桌子
Words remembered	书, 电脑; partial recall of 拿起 — **2/5 clearly remembered**
Task 1 first attempt	No
Wrong selections	Book, Desk
Meaning of 找到电脑	Unsure — knew it involved the computer but did not remember 找到
Meaning of 拿起书	“Pick up the book.”
Task 2 interaction	Clicked first, then experimented with Grab
Most confusing part	Action words and lack of feedback after selections
Most enjoyable part	Picking up the book in VR
Suggested improvement	Show action animations; make Chinese more visually dominant; add feedback/audio
Ease	3/5
Enjoyment	4/5
Key observation	P3 relied heavily on English and initially did not realise objects were selectable. They eventually completed Task 1 but later said they had been guessing, showing that task completion did not necessarily mean language comprehension.

Appendix D — Participant 4 Raw Data

Item	Result / Response
Chinese experience	Basic / Beginner
VR experience	Some
VR confidence	4/5
Vocabulary Learning	1 min 39 sec
Explore Room	43 sec
Task 1	10 sec
Task 2	16 sec
Core task total	**2 min 48 sec**
Difficult words	找到, 拿起
Words remembered	书, 电脑, 找到, 拿起 — **4/5**
Task 1 first attempt	Yes
Meaning of 找到电脑	“Find the computer.”
Meaning of 拿起书	“Pick up the book.”
Task 2 interaction	Grabbed successfully
Most confusing part	No indication that Task 1 had finished
Most enjoyable part	Using the controller to interact with objects
Suggested improvement	Give feedback after every action and allow pronunciation replay

Ease	4/5
Enjoyment	4/5
Key observation	P4 already knew some concrete nouns but still found verbs harder. After correctly selecting the computer, they asked “Did I do it correctly?” because there was no feedback. The broken sequence, rather than Chinese comprehension, was the main problem.

Appendix E — Participant 5 Raw Data

Item	Result / Response
Chinese experience	Little / Basic
VR experience	Little
VR confidence	3/5
Vocabulary Learning	1 min 2 sec
Explore Room	58 sec
Task 1	16 sec
Task 2	25 sec
Core task total	**2 min 41 sec**
Difficult word	拿起
Words remembered	书, 桌子, 电脑, 找到, 拿起 — **5/5 with some hesitation**
Task 1 first attempt	Yes
Meaning of 找到电脑	“Find the computer.”
Meaning of 拿起书	“Pick up the book.” — understood after task
Task 2 interaction	Clicked first, then successfully grabbed
Most confusing part	Thought Task 1 was broken because nothing happened
Most enjoyable part	Learning 拿起 by actually picking something up
Suggested improvement	Add task feedback, consistent room cards and pronunciation replay
Ease	3/5
Enjoyment	4/5
Key observation	P5 relied more on Pinyin than English. After receiving no response in Task 1, they selected the computer again. During Task 2, physically grabbing the book helped them connect 拿起 with its meaning.